

COMP 1111 Final Makeup

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I. QUESTION (20 POINTS)

Write a Java method, named **copyN**, that takes an integer array, A , and three integers, $start$, end and n , as arguments, where $start, end, n \leq 0$. **copyN** copies all elements indexed between $start$ and end of the array A n times and returns the resulting array back.

$A = \{2, 4, 5, 6\}$ $start = 1$, $end = 2$, $n = 4$

$\{4, 5, 4, 5, 4, 5, 4, 5\}$

$A = \{3, 1, 2\}$ $start = 0$, $end = 2$, $n = 3$.

$\{3, 1, 2, 3, 1, 2, 3, 1, 2\}$

II. QUESTION (20 POINTS)

Write a Java method **counts**, that takes an integer array and counts the occurrences of each member of the array and returns those occurrences as an array. You can assume the elements of the array are non-negative and less than 100.

Write another Java method **minCount**, that takes an integer array and returns the least frequent member. You must use **counts** method in this second method.

$A = \{6, 2, 4, 5, 6, 2, 3, 6, 8, 4, 6, 8, 6\}$
Returns 3

$A = \{2, 4, 2, 1, 1, 1, 2, 1\}$
Returns 4.

III. QUESTION (20 POINTS)

Write a Java method **sortAlternate**, which takes a 2d array A as argument and sorts the rows of the arrays alternately. First half of the array is sorted in increasing order, second half in decreasing order. Write helper methods

sortIncreasingOrder(int[][] A, int rowNo)

sortDecreasingOrder(int[][] A, int rowNo),

which sorts the row of the 2d-array A with index $rowNo$ in increasing and decreasing order respectively. Then use those helper methods in **sortAlternate**.

$$A =$$

3	7	12	1
1	8	4	6
6	14	0	2
8	2	7	12

$$A' =$$

1	3	7	12
1	4	6	8
14	6	2	0
12	8	7	2

IV. QUESTION (20 POINTS)

Suppose that we are modelling AmiralBattı game. Write the Java method **int numberOfNonHits(boolean[][] shots, boolean[][] shipParts)**, which counts the number of ship parts that are not hit. In this function; $shots[i][j]$ is true, if we have shot the cell in the i 'th row and j 'th column; false otherwise. Yet in this function; $shipParts[i][j]$ is true, if there is a ship part in the cell in the i 'th row and j 'th column.

$$shots =$$

true	true	false	false
false	false	false	false
true	true	false	false
false	false	false	false

$$shipParts =$$

true	false	true	true
false	false	false	false
false	true	true	true
false	false	false	false

numberOfNonHits returns 4.

V. QUESTION (20 POINTS)

Write a method **paste**, which takes two arrays A and B and two values k, m as arguments, and returns a new array which inserts the array B at positions k and m in the array A and returns the resulting array.

$A = \{2, 4, 5, 7, 8\}$, $B = \{8, 10, 12\}$
 $k = 1$, $m = 3$.

$\{2, 8, 10, 12, 4, 5, 8, 10, 12, 7, 8\}$

$A = \{3, 4, 8, 9\}$, $B = \{7, 10, 12, 20, 24\}$
 $k = 0$, $m = 3$.

$\{7, 10, 12, 20, 24, 3, 4, 8, 7, 10, 12, 20, 24, 9\}$